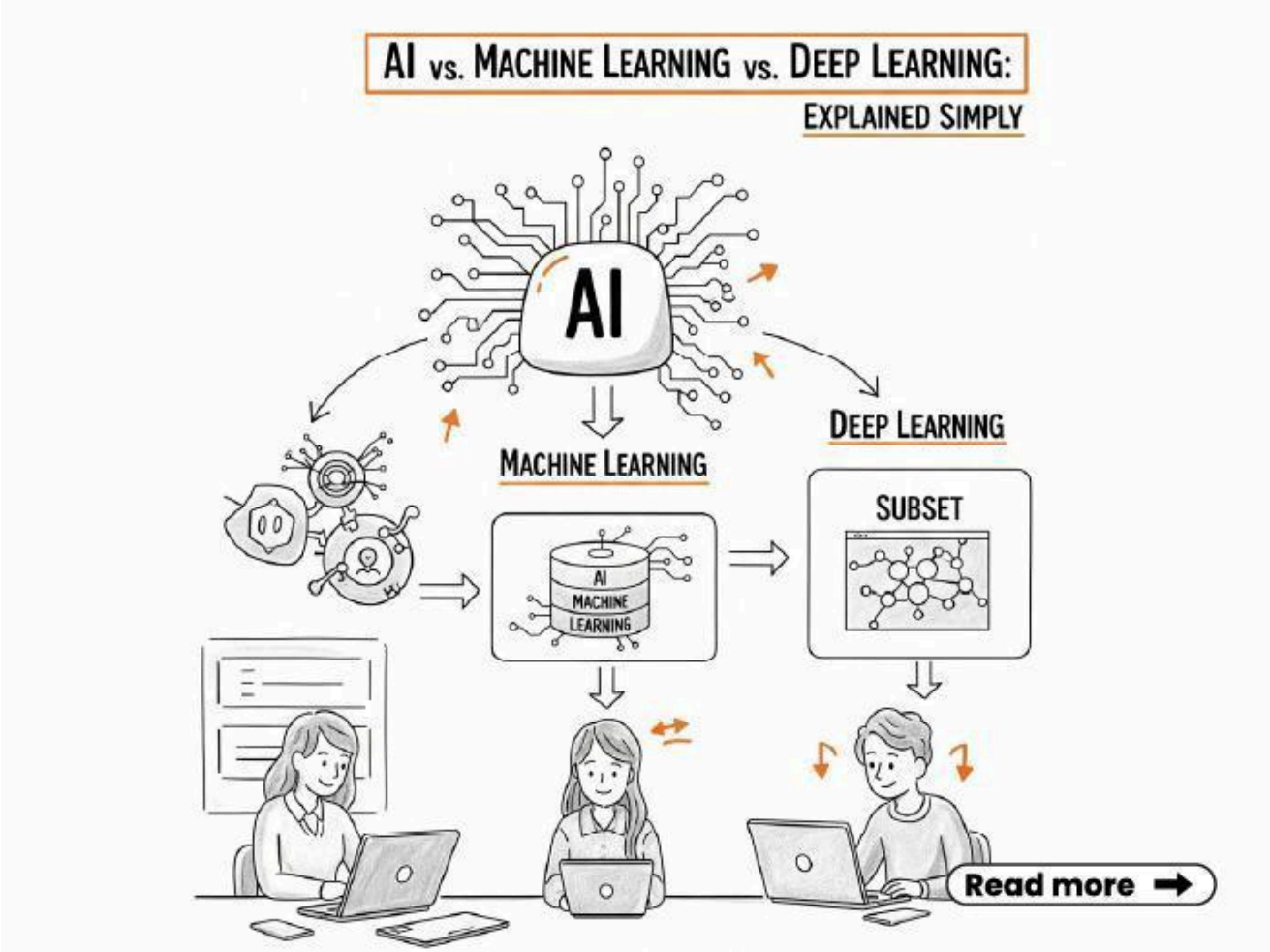
Machine Learning

Research Assignment

Bright Learn

Data Science April Intake



Name: Charlotte Turya

Date: 2 November 2026

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Part A: What is machine learning?

A1. What is machine learning?

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and use them to make predictions or decisions. In machine learning, an algorithm learns patterns from examples and uses the resulting model to process new data. Machine learning provides computers the ability to learn from data and improve performance.

For example, a bank can train a model using historical transactions labelled as fraudulent or legitimate. The model can then estimate whether new transactions are potentially fraudulent. Its predictions can support human decision-making and fraud prevention.

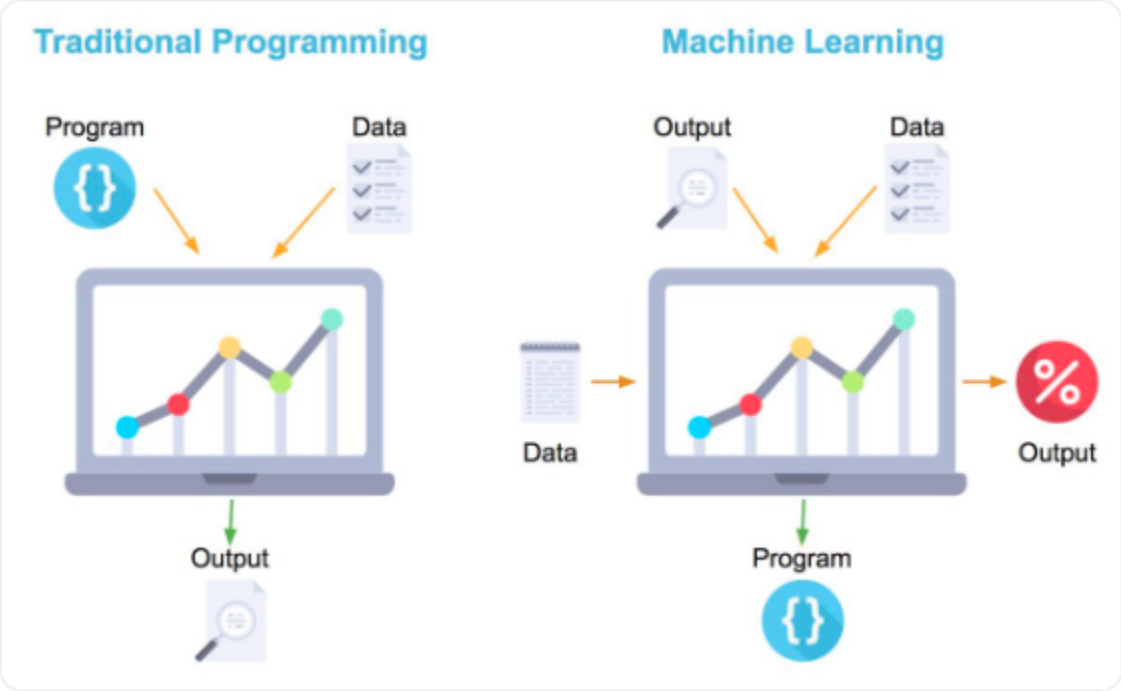


Image Source: https://efxa.org/2021/04/17/traditional-programming-and-machine-learning/
Source: https://developers.google.com/machine-learning/intro-to-ml/what-is-ml

A2. How does machine learning relate to AI, deep learning and data science?

AI is the overall goal of making machines intelligent, Machine Learning enables machines to learn from data, Deep Learning uses more complex neural networks to learn patterns, and Data Science uses data and analytical techniques to solve problems and support decision-making.

Machine Learning (ML) is a subset of Artificial Intelligence (AI) and is closely related to Deep Learning and Data Science. Although these fields overlap, they have different purposes.

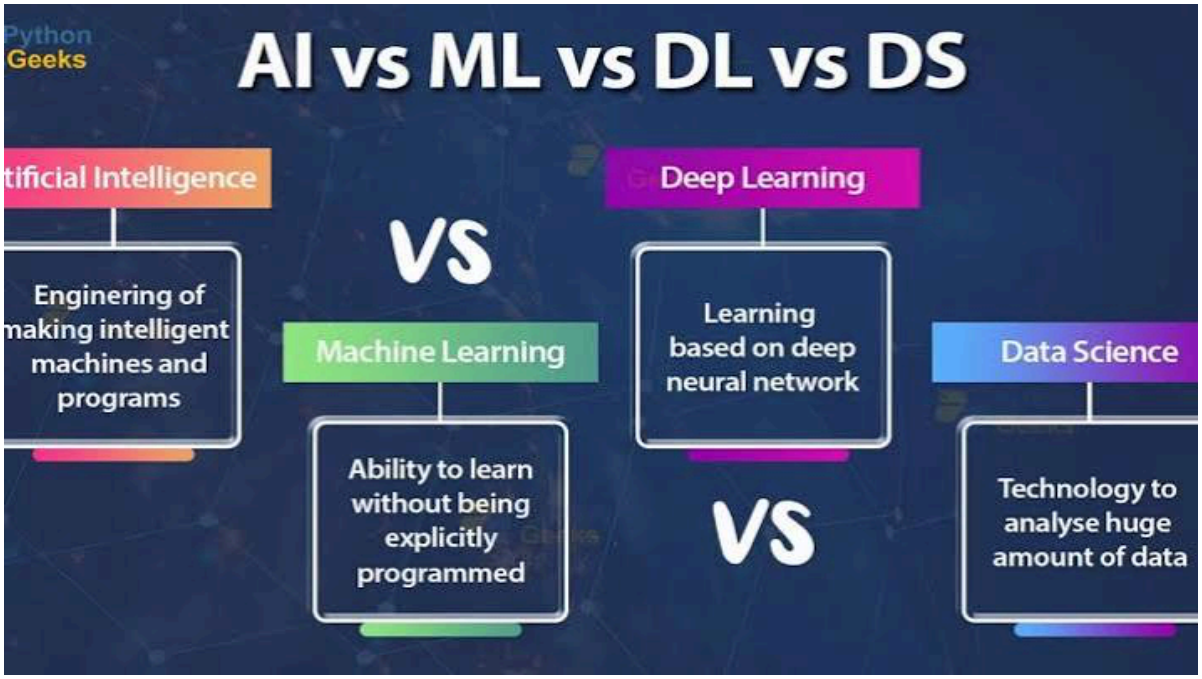


Image Source: https://pythongeeks.org/ai-vs-data-science-vs-deep-learning-vs-ml/

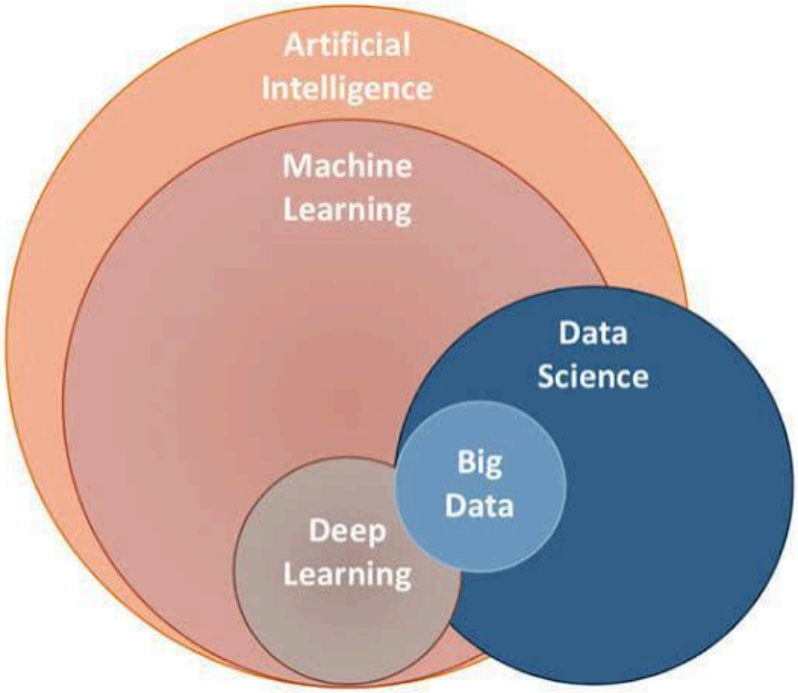


Image source: https://ai.plainenglish.io/data-science-vs-artificial-intelligence-vs-machine-learning-vs-deep-learning-50d3718d51e5

A3. Core terms: dataset, feature, label (target), model, training, prediction

A dataset is a collection of observations, such as records of learners' academic performance. A feature is an input variable used by a model, such as study hours or a previous test mark. A label or target is the outcome that the model must predict, such as the learner's final mark. Training is the process through which an algorithm learns patterns from example data. The resulting model represents those learned patterns. A prediction is the output produced when the trained model receives new input data.

Learner	Study hours per week	Previous test mark	Final mark
A	2	45	48

B	5	60	65
C	8	75	82
D	6	68	72

Imagine that a school wants to predict learners' final marks using their previous academic results.

1. Dataset

The complete collection of data used for analysis or machine learning. Here, the dataset contains learners' study hours, previous marks and final marks.

2. Feature

An input variable the model uses to make a prediction. Study hours and previous test marks are features.

3. Label (target)

The answer the model is being trained to predict. In this example, the final mark is the target.

4. Model

The learned relationship between the input features and the target. After training, the model can estimate a learner's final mark.

5. Training

The process of using historical examples to help an algorithm learn patterns and adjust its model.

6. Prediction

The output generated when a trained model receives input data. For example, it might predict a final mark of 70 for a new learner.

A4. Three real-world uses of machine learning in South Africa or elsewhere in Africa

Machine learning has several potential applications in Africa.

1. Mobile banking fraud detection

- ➔ **Real-World Examples: Safaricom (M-Pesa)** operating in Kenya and **Nedbank** within South Africa.
- ➔ **Model Inputs:** Transaction amounts, velocity, timestamps, geographic locations, and historical customer spending patterns.
- ➔ **Target Label:** Determining whether a specific payment is legitimate or fraudulent.
- ➔ **Practical Impact:** Protects clients and banking systems by identifying unauthorized transactions before financial loss occurs.
- ➔ **How it works in reality:** Across East Africa, Safaricom M-Pesa applies graph neural network architectures to analyse account connections and detect suspicious behavior instantly, lowering platform fraud by 87%. In South Africa, Nedbank's Risk Intelligence Centre deploys machine-learning algorithms across integrated infrastructure to uncover fraudulent networks and money laundering activities before payments complete. Safaricom provides an example of how artificial intelligence can support fraud prevention in digital financial services. In its 2025 Sustainable Business Report, the company reported an 87% reduction in fraud through the use of AI to identify fraudsters involved in social engineering. Safaricom also developed an AI-powered fraud detection framework that includes fraud risk scoring, registration intent, IMEI similarity and identity theft detection (Safaricom PLC, 2025).

  Source:https://kenyanwallstreet.com/how-safaricom-is-leveraging-ai-to-bolster-m-pesa-security-and-efficiency

2. Crop disease detection

- ➔ **African Deployment:** The **Crop Disease Detection App** developed by the **Responsible AI Lab (RAIL)**in Ghana.
- ➔ **Input Data:** Photographs of leaves, crop images, and weather data.
- ➔ **Prediction:** The likely crop disease or signs of infection.
- ➔ **Beneficiaries:** Farmers and agricultural advisers, who can respond earlier to potential crop losses.

- ➔ **How it works in reality:** Smallholder farmers use a smartphone application to capture images of infected plant leaves. A computer vision model classifies up to 23 distinct crop diseases locally. To ensure localized access, the app reads out the predicted disease and immediate cultural or chemical treatment steps via automated voice messages in local languages, including Asante Twi, Ewe, and Hausa.

Source: https://www.ai4d.ai/innovations/crop-disease-detection-app

3. Telecommunications network fault prediction

- ➔ **African Deployment: MTN Group** (deployed across 19 African markets) and **Vodacom** in South Africa.
- ➔ **Input Data:** Network performance measurements, traffic telemetry, error logs, and equipment sensor readings.
- ➔ **Prediction:** Whether equipment or a network component may fail.
- ➔ **Beneficiaries:** Telecommunications providers and their customers, through better maintenance planning and fewer service interruptions.
- ➔ **How it works in reality:** MTN Group's ADAM Platform processes close to 4 trillion data records monthly using Vertex AI across its African footprint to break down data silos and optimize network performance. Meanwhile, Vodacom's core network teams utilize an online learning framework that streams infrastructure counters, traffic anomalies, and APN statistics to predict service-affecting failures in mobile core infrastructure before they spark a region-wide outage.MTN Group provides another example of how large organisations use cloud-based data platforms to support analytics. According to Google South Africa's interim country director, MTN uses Google Cloud solutions to improve customer experience, with the platform enabling the processing of approximately four trillion data records monthly. This demonstrates how cloud computing and data analytics can help organisations manage large volumes of information and generate business insights (Malinga, 2024).

Source:https://www.facebook.com/SafaricomNews/videos/over-70-ai-and-machine-learning-models-are-already-live-in-production-quietly-pr/1719490336031580/

A5. The machine learning workflow

The machine learning workflow begins by defining the problem and deciding what the model should predict. Relevant data is then collected, explored and prepared by addressing missing values, duplicates, incorrect data types and categorical variables. A suitable algorithm is

selected and trained using the prepared training data. The model is evaluated on separate test data to estimate its performance on unseen examples. If its performance is satisfactory, it can be deployed and monitored. Data preparation often takes considerable time because real-world datasets are not always complete, consistent or immediately suitable for modelling.

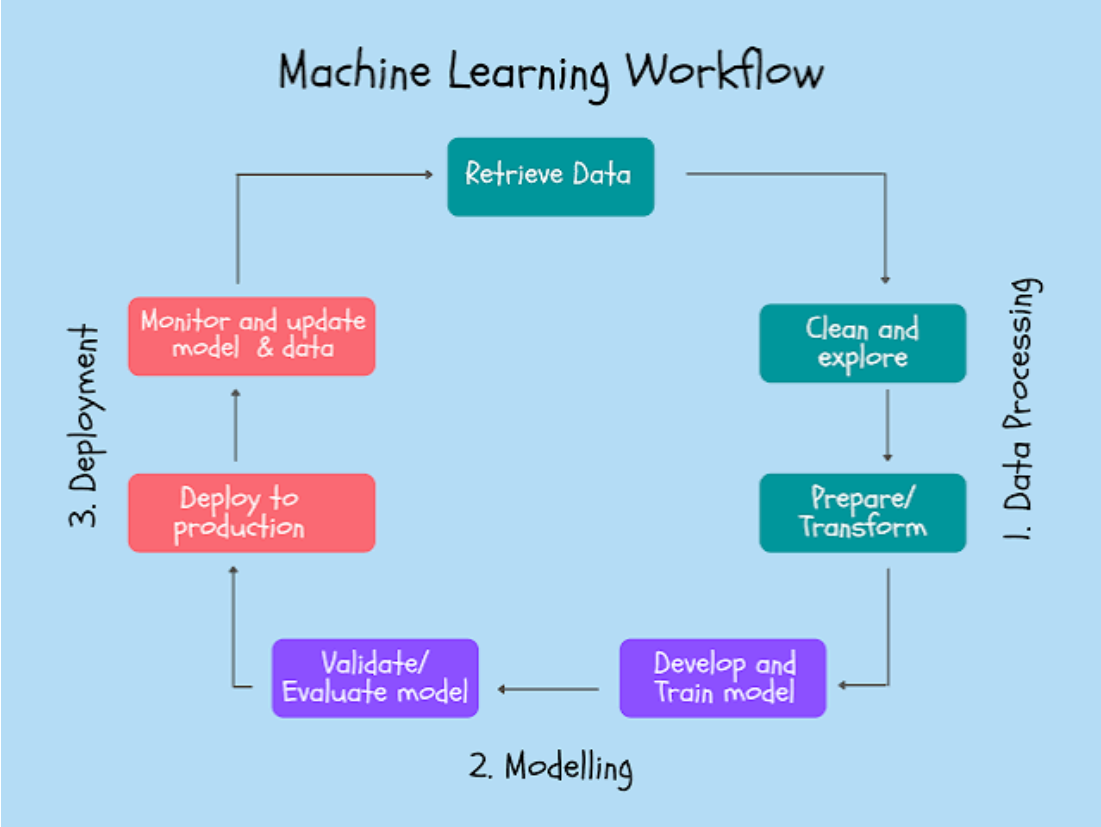


Image Source: https://medium.com/data-science/the-machine-learning-workflow-explained-557abf882079

Why does data preparation take so much time?

Real-world data may contain missing values, duplicated records, inconsistent formats, incorrect entries and categories stored as text. The data must be checked and transformed into a form the algorithm can use. Poor preparation can lead to unreliable predictions, even when an appropriate algorithm is selected.

Data preparation also needs care to avoid data leakage — accidentally allowing information from the test data to influence model training.

Source: scikit-learn developers (n.d.) Getting Started. Available at: scikit-learn official documentation.

Part B: Supervised vs unsupervised learning

B1. What is supervised learning?

Supervised learning is a machine learning approach in which an algorithm learns from labelled data containing both input features and known target values. For example, an email filter can learn from messages labelled as spam or not spam and use those patterns to classify new messages.

B2. What is unsupervised learning?

Unsupervised learning uses data without predefined target labels to discover patterns, similarities or groups. For example, a bank can group customers according to their spending habits without defining the groups beforehand.

B3. Comparison table

	Supervised learning	**Unsupervised learning**
Goal	Predict a known target or answer	Discover hidden patterns or structure
Type of data needed	Features and labelled target values	Features without predefined target labels
Typical tasks	Classification and regression	Clustering and dimensionality reduction
Example algorithms	Linear Regression, Logistic Regression, Decision Tree Classifier	K-Means, PCA
How success is measured	Compare predictions with known answers using	Use measures such as silhouette score, or assess

	Supervised learning	Unsupervised learning
	metrics such as accuracy, F1-score or MAE	whether the discovered groups are useful and meaningful
Real-world example	Predict whether a transaction is fraudulent using historically labelled transactions	Group customers by similar spending habits without predefined group labels

B4. Classification and regression

Both are types of supervised learning, but they predict different kinds of answers.

Classification is a supervised learning task that predicts a category, such as whether an SMS is spam or not spam. Example:Predicting whether a learner will pass or fail.

Regression is a supervised learning task that predicts a numerical value. Example: A house's selling price.

One useful distinction: a classification model may internally calculate a probability, such as a 75% probability of fraud, but its classification output is a category. A regression model estimates a numerical quantity.

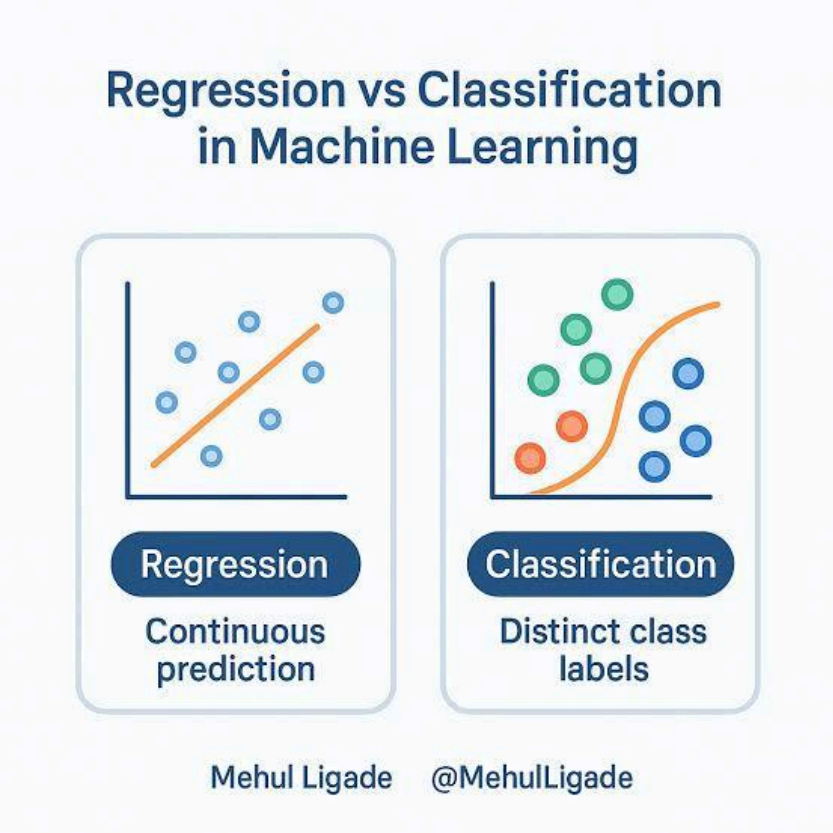


Image source:
https://pub.towardsai.net/regression-vs-classification-in-machine-learning-why-most-beginners-get-this-wrong-m004-7e01b32602ec?gi=7104ba230cb5

B5. Clustering and dimensionality reduction

Clustering is an unsupervised learning technique that groups observations according to their similarities. For example, K-Means can group retail customers with similar purchasing behaviour.

Dimensionality reduction decreases the number of variables used to represent a dataset while retaining important patterns. For example, PCA can reduce hundreds of related measurements to a smaller set of components, making complex data easier to explore.

	Clustering	Dimensionality reduction
What it does	Groups similar observations together without requiring predefined group labels	Reduces the number of variables used to represent data while retaining as much relevant structure or information as possible
Example	A retailer groups customers according to their purchase patterns	Simplifying a dataset containing hundreds of measurements so it can be explored more easily
Algorithm	K-Means	Principal Component Analysis (PCA)
scikit-learn module	sklearn.cluster	sklearn.decomposition

B6. Classify the five scenarios

Scenario	Supervised or unsupervised?	Task type	Why (1–2 sentences)
(a) House price from size and suburb	Supervised	Regression	Historical house prices provide known numerical targets.
(b) Grouping customers by spending habits	Unsupervised	Clustering	The algorithm discovers groups based on similarities

Scenario	Supervised or unsupervised?	Task type	Why (1–2 sentences)
			in spending behaviours.
(c) Deciding whether an SMS is spam	Supervised	Classification	Training messages are labeled spam vs not spam.
(d) Predicting a learner's final mark from term marks	Supervised	Regression	The target is a numerical mark learned from previous examples.
(e) Finding groups of products with similar sales patterns	Unsupervised	Clustering	The algorithm groups products based on similarities in their sales data without predefined group labels.

How to reason through these questions

Ask yourself two questions:

1. Are correct answers provided in the training data? If yes, supervised learning; if not, potentially unsupervised learning.
2. What kind of result is required? A category means classification, a numerical value means regression, and groups of similar observations mean clustering.

B7. Scikit-learn classes

Class	Module	Supervised or unsupervised?	What it is used for
LinearRegression	sklearn.linear_model	Supervised	Regression: predicting a number
LogisticRegression	sklearn.linear_model	Supervised	Classification: predicting a category
DecisionTreeClassifier	sklearn.tree	Supervised	Classification: predicting a category
KNeighborsClassifier	sklearn.neighbors	Supervised	Classification: predicting a category
KMeans	sklearn.cluster	Unsupervised	Clustering: grouping similar observations
PCA	sklearn.decomposition	Unsupervised	Dimensionality reduction: fewer variables

Semi-Supervised Learning

Think of this as a middle ground where you don't have enough time or money to label a massive dataset. Instead of manually tagging thousands of items, you label a tiny fraction of them yourself. The algorithm studies that small, labeled batch to understand the basics, and then it looks at the mountain of remaining, unlabeled data to fill in the blanks on its own.

- Example**:** Imagine trying to sort thousands of **medical scans** for a rare disease. A doctor only has time to manually tag 50 of them. The system looks at those 50 tagged scans to learn what the disease looks like, and then it applies that knowledge to sort the thousands of remaining, untagged images.

Reinforcement Learning

This approach is basically learning by trial and error, exactly how a human or a pet learns. You drop an AI "agent" into an environment with a specific goal, but you don't give it any instructions on how to achieve it. Every time it makes a move that gets it closer to the goal, it gets a digital "reward" (like a treat). Every time it makes a mistake, it gets a penalty. It repeats this loop millions of times until it figures out the perfect strategy.

- Example: Think of teaching a **self-driving car** how to navigate a track. If it stays in the lane, it gets points. If it hits a guardrail or drives off the road, it loses points. After practicing on a simulator over and over, it learns the safest and fastest way to drive.

Part C: Preparing the data

C1. Explore and clean

When opening a new dataset, you should understand its structure and quality before doing any analysis or machine learning.

These are the most important checks:

What to check	Pandas function	Purpose
Dataset size	df.shape	Shows rows and columns
First few records	df.head()	Gives a preview of the data
Data types	df.info()	Identifies numerical and text columns
Statistical summary	df.describe()	Shows averages, minimums, maximums and other statistics
Missing values	df.isnull().sum()	Counts missing values in each column

Duplicate records	df.duplicated().sum()	Identifies the total number of duplicate records
Incorrect data types	df.astype()	Changes the data type

C2. Handle missing values

Missing values can be handled by either removing the affected data or replacing the missing values with suitable alternatives. In Python, pandas provides the dropna() function to remove rows or columns containing missing values. This approach is appropriate when only a small number of records are affected or when a column contains too much missing information to be useful.

However, dropping rows can be problematic, particularly when working with a small dataset, because it reduces the amount of information available for analysis and model training.

Another approach is to replace missing values using fillna() or Scikit-learn's SimpleImputer. Common imputation methods include the mean, median, mode and a constant value. The mean can be used for numerical data without extreme outliers, while the median is often more appropriate when outliers are present because it is less affected by unusually high or low values. The mode is useful for categorical data because it replaces missing values with the most frequently occurring category.

In my Titanic dataset, I used the median to replace missing numerical values and the most frequent value to replace missing categorical values. I excluded the Cabin column from my selected model features because approximately 77% of its values were missing. This approach allowed me to retain useful records while addressing missing information.

C3. Encode categorical data

Machine-learning algorithms generally require numerical data because they use mathematical calculations to identify patterns and relationships. Categorical data, such as gender, location or

customer satisfaction levels, is often stored as text. Encoding is therefore used to convert these categories into numerical representations that machine-learning models can process.

There are different encoding techniques depending on the type of categorical variable.

Label encoding assigns a numerical value to each category. It is commonly used to encode target labels. However, using it on nominal input features can incorrectly suggest that one category has a higher value or ranking than another.

Ordinal encoding is suitable for categorical variables that have a natural order or ranking. For example, customer satisfaction levels such as poor, average, good and excellent can be represented by values from 1 to 4. This preserves the order of the categories.

One-hot encoding is suitable for nominal categories that do not have a natural order. It creates a separate binary column for each category, using 1 to indicate the presence of a category and 0 to indicate its absence. In my Titanic dataset, I used one-hot encoding for the Sex and Embarked columns.

One limitation of one-hot encoding is that it can create a large number of additional columns when a feature contains many unique categories. This is known as high cardinality. It can increase memory usage, make the dataset more complex and increase the computational resources required to train a model.

The choice of encoding method therefore depends on whether categories have a natural order and how many unique categories are present.

C4. Scale numeric features

Feature scaling is a data-preprocessing technique used to bring numerical features onto comparable scales. It is important because some machine-learning algorithms are sensitive to differences in the ranges of numerical values. For example, in a dataset containing Age and Fare, the Fare values may be much larger than the Age values. Without scaling, features with larger values can have a greater influence on certain algorithms.

Two common scaling methods are standardisation and normalisation. Standardisation, which can be performed using Scikit-learn's StandardScaler, transforms values using the mean and standard deviation. It generally produces data with a mean close to zero and a standard deviation close to one. Normalisation, which can be performed using MinMaxScaler, rescales values into a specified range, commonly between zero and one.

Another method is RobustScaler, which uses the median and interquartile range. It is less sensitive to extreme values and can be useful when a dataset contains outliers.

Feature scaling is particularly important for algorithms such as K-Nearest Neighbours (KNN), Logistic Regression and Support Vector Machines (SVM), because their calculations are affected by feature magnitudes. Decision Trees and Random Forests generally do not require scaling because they make decisions using feature thresholds rather than distance calculations.

In my Titanic notebook, I used StandardScaler to scale Age, Fare, Pclass, SibSp, Parch and the engineered FamilySize feature. I fitted the scaler on the training data and applied the same transformation to the test data to prevent data leakage.

C5. Split into training and test sets

A dataset is divided into training and testing sets to evaluate whether a machine-learning model can make accurate predictions using data it has not previously seen. The training set is used to teach the model and identify patterns, while the testing set is used to assess its performance on unseen data.

A common approach is an 80/20 split, where 80% of the dataset is allocated to training and 20% to testing. Scikit-learn provides the train_test_split() function to perform this division.

The random_state parameter ensures that the split can be reproduced when the code is run again. The stratify parameter can be used to maintain approximately the same distribution of target classes in both sets. This is particularly useful when working with classification datasets where the target categories may be imbalanced.

Data leakage occurs when information from the testing data influences the model training or preprocessing process. This can result in misleading performance measurements. To prevent leakage, preprocessing methods such as imputation and scaling should be fitted on the training data only.

In Scikit-learn, fit() learns information from the data, transform() applies the learned transformation, and fit_transform()performs both operations. The correct approach is to use fit_transform() on the training data and transform() on the testing data.

In my Titanic notebook, I used an 80/20 split with random_state=42 and stratify=y. I then fitted the imputers, encoder and scaler using the training data and applied the learned transformations to the testing data.

C6. Identifying and Handling Outliers

Outliers are values that differ significantly from most observations in a dataset. They can affect statistical analysis and machine-learning model performance, particularly when models are sensitive to extreme values.

In Python, several tools can be used to identify and handle outliers:

1. df.quantile()

The df.quantile() function calculates the quantiles of numerical columns. It can be used to determine the first quartile (Q1) and third quartile (Q3), which are required for the Interquartile Range (IQR) method.

```
Q1 = df['Fare'].quantile(0.25)

Q3 = df['Fare'].quantile(0.75)

IQR = Q3 - Q1
```

The IQR represents the middle 50% of the data. Values below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR are commonly flagged as potential outliers.

2. df.clip()

The df.clip() function limits values to specified lower and upper boundaries. This can be used to cap extreme values rather than removing the observations.

```
df['Fare'] = df['Fare'].clip(lower_bound, upper_bound)
```

This example limits Fare values to the specified boundaries. The lower and upper limits must first be calculated using the chosen outlier detection method.

3. seaborn.boxplot()

A boxplot provides a visual way to identify potential outliers. It displays the median, quartiles and values that fall outside the whiskers.

```
import seaborn as sns

import matplotlib.pyplot as plt

sns.boxplot(x=df['Fare'])

plt.show()
```

In my Titanic dataset, the IQR method identified 29 potential outliers in Age and 91 in Fare. These observations were retained because extreme values are not necessarily incorrect, and removing them could result in losing useful information.

C7. Engineer features

Feature engineering involves creating or transforming variables to help a machine-learning model identify useful patterns in the data.

1. Converting dates using pd.to_datetime()

The pd.to_datetime() function converts a column into a datetime format. This allows dates to be analysed and used to create additional features.

```
df['Date'] = pd.to_datetime(df['Date'])
```

2. Extracting the month using .dt.month

Once a column has been converted to datetime format, the .dt.month accessor can extract the month as a numerical value.

```
df['Month'] = df['Date'].dt.month
```

This can help identify seasonal patterns, such as changes in sales or customer activity across different months.

3. Correlation analysis using df.corr()

The df.corr() function calculates the correlation between numerical variables. Correlation values range from -1 to +1. Positive values indicate that two variables tend to increase together, while negative values indicate an inverse relationship. Values closer to zero indicate a weaker linear relationship.

In my Titanic dataset, correlation analysis produced the following results:

- Pclass and Survival: -0.348
- Fare and Survival: +0.275
- Age and Survival: -0.073
- FamilySize and Survival: +0.027
- FamilySize and SibSp: +0.874
- FamilySize and Parch: +0.789

These results suggest that the passenger class had a negative relationship with survival, while Fare had a positive relationship. FamilySize had a very weak positive correlation with survival. Its stronger correlations with SibSp and Parch are expected because FamilySize was calculated using these two variables. Correlation does not establish causation, but it helps identify relationships that may be useful during exploratory data analysis and feature selection.

Notebook summary

For the practical component, I used the Titanic training dataset downloaded from Kaggle. The dataset contained 891 passenger records and 12 original columns. I used Python, pandas and Scikit-learn to explore, clean and prepare the data for machine-learning purposes.

I first inspected the dataset using pandas functions to understand its structure, dimensions, data types and missing values. The initial investigation identified missing values in the Age, Cabin and Embarked columns. I selected Survived as the target variable because the objective was to examine passenger survival. The input features included Pclass, Sex, Age, SibSp, Parch, Fare and Embarked. I excluded Cabin from the selected features because approximately 77% of its values were missing.

I divided the data into training and testing sets using an 80/20 split. This produced 712 training records and 179 testing records. I used random_state=42 to make the split reproducible and stratify=y to maintain the proportion of survival classes in both sets.

To handle missing values, I used Scikit-learn's SimpleImputer. The median strategy was applied to numerical features, while the most frequent value was used for categorical features. I then applied one-hot encoding to the Sex and Embarked columns using OneHotEncoder, converting the categorical values into numerical columns that could be processed by machine-learning algorithms.

I used StandardScaler to standardise the numerical features, including Age, Fare, Pclass, SibSp, Parch and FamilySize. The imputer, encoder and scaler were fitted on the training data and then applied to the testing data. This helped prevent data leakage by ensuring that information from the testing set was not used to learn the preprocessing parameters.

I also investigated potential outliers in the Age and Fare columns using the Interquartile Range (IQR) method. The analysis identified 29 potential outliers in Age and 91 in Fare. These observations were retained because unusual values are not necessarily incorrect and should be investigated before being removed.

I also used boxplots to visually examine the distribution of numerical features and identify potential outliers in the Age and Fare columns. The boxplots helped me observe the median, spread of the data and values outside the whiskers. I investigated the potential outliers further using the Interquartile Range (IQR) method. This identified 29 potential outliers in Age and 91 in Fare. I retained these observations because unusual values are not necessarily errors and may represent genuine passenger information.

I performed feature engineering by creating a new column called FamilySize. This was calculated by adding SibSp and Parch, plus one to include the passenger. This feature provides additional information about the size of each passenger's family group aboard the ship.

Finally, I created a correlation matrix to examine the relationships between the numerical features and the target variable, Survived. Pclass had the strongest relationship with survival, with a correlation of -0.348, followed by Fare at 0.275. FamilySize had a weak correlation with survival of 0.027. I also found that FamilySize was strongly correlated with SibSp (0.874) and Parch (0.789), which is expected because these variables were used to calculate FamilySize. This exercise helped me understand relationships between features and the survival outcome, while recognising that correlation does not establish causation.

After preprocessing, the final training feature matrix contained 712 records and 11 features, while the testing feature matrix contained 179 records and 11 features. The dataset was therefore cleaned, encoded and scaled, making it ready for use in a machine-learning model.

The practical exercise demonstrated the importance of data inspection, handling missing values, categorical encoding, feature scaling, outlier investigation, feature engineering and correct training/testing data preparation.

Part D: Further research

D1. Overfitting and Underfitting

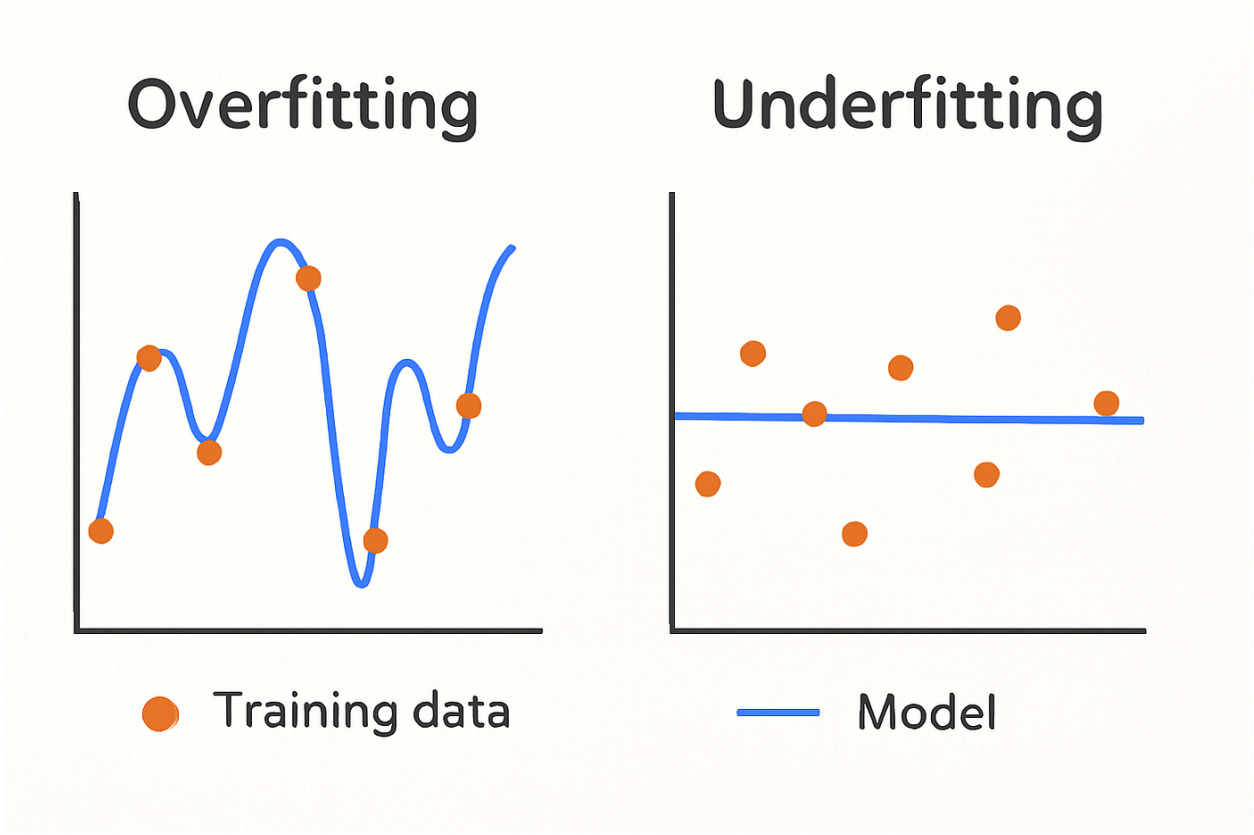
Overfitting and underfitting are two common problems that affect the performance of machine-learning models.

Overfitting occurs when a model learns the training data too closely, including noise and unnecessary details. As a result, it performs very well on the training data but poorly on new, unseen data. For example, a Titanic survival prediction model may memorise patterns from its training passengers instead of learning relationships that can be applied to other passengers.

Underfitting occurs when a model is too simple to identify important patterns in the dataset. It performs poorly on both training and testing data because it has not learned enough from the available information.

A train/test split helps identify these problems by allowing the model's performance to be compared across training and unseen testing data. Overfitting is indicated by strong training performance and weak testing performance, while underfitting results in poor performance on both sets. A good fit occurs when the model performs reasonably well on both.

A graph comparing model complexity with training and testing error can illustrate these differences. Underfitting occurs when both errors are high, while overfitting occurs when training error is low but testing error increases. The aim is to find a model that learns meaningful patterns without memorising the training data.



A graph comparing model complexity with training and testing error can illustrate these differences. Underfitting occurs when both errors are high, while overfitting occurs when training error is low but testing error increases. The aim is to find a model that learns meaningful patterns without memorising the training data (Scikit-learn Developers, n.d.).

Image source:
https://medium.com/@Monica-Ashok/day-11-overfitting-vs-underfitting-the-art-and-science-of-building-reliable-machine-learning-cee616373046

D2. Measuring a Model

Machine-learning models are evaluated using different metrics depending on whether they perform classification or regression.

For classification, a confusion matrix shows the number of correct and incorrect predictions. It includes true positives, true negatives, false positives and false negatives. Accuracy measures the proportion of all predictions that are correct. Precision measures how many of the cases predicted as positive are actually positive, while recall measures how many of the actual positive cases the model successfully identifies. The F1-score combines precision and recall into a single measure, balancing both.

For regression, Mean Absolute Error (MAE) measures the average absolute difference between predicted and actual values. Root Mean Squared Error (RMSE) also measures prediction error but penalises larger errors more heavily. R-squared (R^2) indicates how much variation in the target variable is explained by the model (Scikit-learn Developers, n.d.).

Accuracy can be misleading when detecting fraud because fraudulent transactions are often a small proportion of all transactions. If only 1 in 100 transactions is fraudulent, a model that classifies every transaction as legitimate could achieve 99% accuracy while failing to detect any fraud. In this situation, precision and recall are more useful. Recall helps measure how many fraudulent transactions are detected, while precision indicates how many flagged transactions are genuinely fraudulent. The F1-score can help balance these two measures.The F1-score combines precision and recall into a single measure, balancing both (Scikit-learn Developers, n.d.).

Source : Scikit-learn Developers (n.d.). Model evaluation: quantifying the quality of predictions. Available at: https://scikit-learn.org/stable/modules/model_evaluation.html

D3. Cross-validation

Cross-validation is a technique used to evaluate how well a machine-learning model is likely to perform on unseen data. One commonly used method is K-fold cross-validation, where the dataset is divided into K approximately equal sections, known as folds.

For example, in 5-fold cross-validation, the model is trained five times. During each round, four folds are used for training and the remaining fold is used for validation. The validation fold changes in each round until every fold has been used once. The performance scores from all five rounds are then averaged to produce an overall evaluation score.

Cross-validation is generally more reliable than using a single train/test split because the model is evaluated across different portions of the dataset rather than relying on one particular

division. This provides a better indication of how consistently the model performs and reduces the risk of drawing conclusions from an unusually favourable or unfavourable split.

Scikit-learn provides a function called cross_val_score, available from sklearn.model_selection, to perform cross-validation. For example, setting cv=5 requests five-fold cross-validation. The function returns a score for each fold, which can be averaged to assess the model's performance. Preprocessing should be included in a pipeline to help prevent data leakage between training and validation folds (Scikit-learn Developers, n.d.).

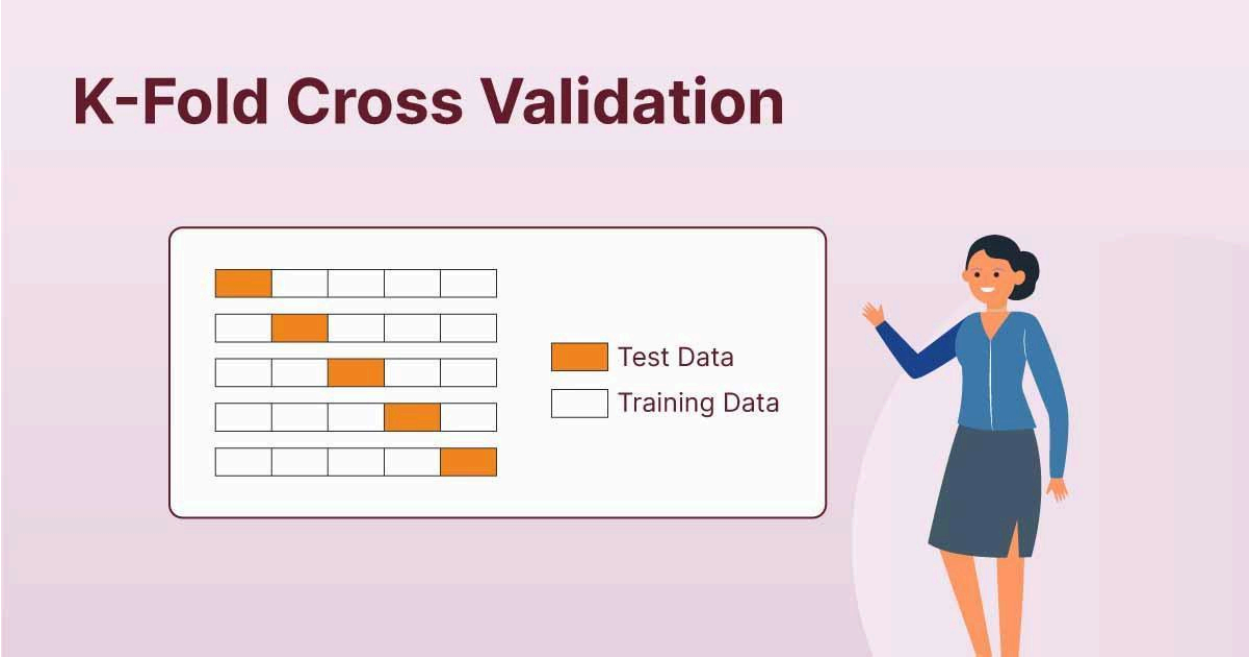


Image source: https://www.shiksha.com/online-courses/articles/k-fold-cross-validation/

Source: Scikit-learn Developers (n.d.). Cross-validation: evaluating estimator performance. Available at: https://scikit-learn.org/stable/modules/cross_validation.html

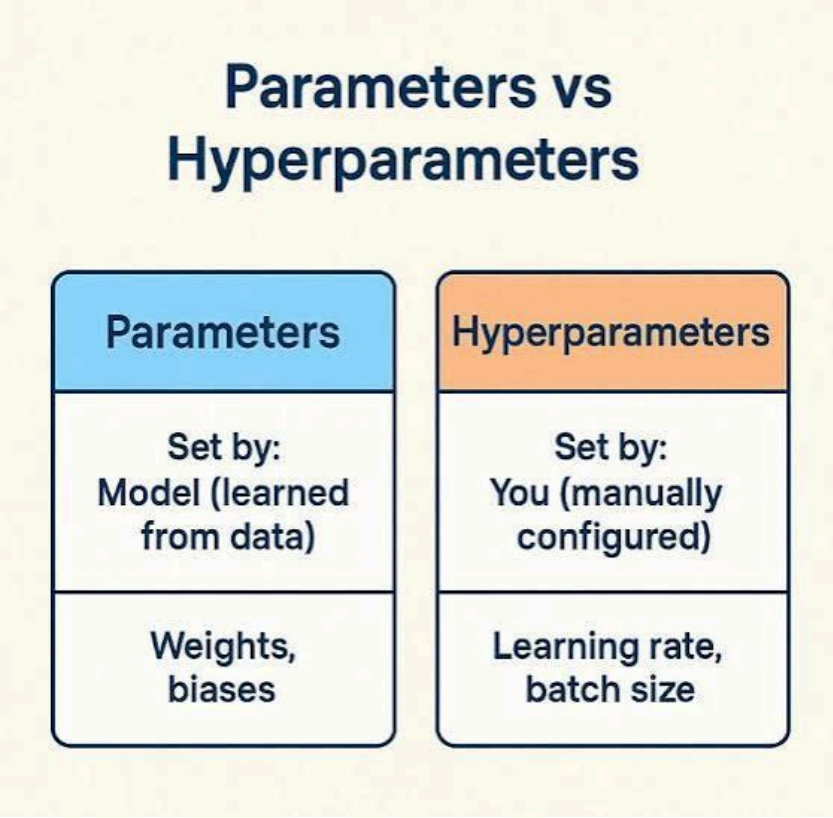
D4. Parameters vs Hyperparameters

Parameters and hyperparameters are both important in machine learning, but they serve different purposes. Parameters are values that a model learns automatically from the training data. For example, a decision tree learns which features to use for splitting the data and the threshold values at which these splits should occur.

Hyperparameters are settings selected before training to control how the model learns. In a decision tree, examples include max_depth, which limits how deep the tree can grow, and min_samples_split, which specifies the minimum number of samples required to split a node.

Scikit-learn provides GridSearchCV to help identify suitable hyperparameter settings. It tests different combinations of values and evaluates their performance using cross-validation. For example, a decision tree could be tested with maximum depths of 2, 4, 6 and 8 to identify which setting performs best (Scikit-learn Developers, n.d.).

For example, if I test four different max_depth values (2, 4, 6 and 8) using 5-fold cross-validation, GridSearchCV will perform 20 model fits. This is because each of the four hyperparameter settings is evaluated across five different folds (4 × 5 = 20). If more hyperparameter combinations are added, the number of model fits increases, making the search more computationally expensive.



Source: Scikit-learn Developers (n.d.). Tuning the hyper-parameters of an estimator. Available at: https://scikit-learn.org/stable/modules/grid_search.html

Image source:
https://www.linkedin.com/posts/shaiju-ms-b9b9b5b5_aiforbeginners-hyperparameters-machinelearning-activity-7409102864985493504-RXJG

D11. Data Protection and POPIA

The Protection of Personal Information Act (POPIA), Act 4 of 2013, is South African legislation that protects personal information and regulates how public and private organisations collect, store, use and share it. It establishes eight conditions for lawful processing, including accountability, purpose specification, information quality, openness and security safeguards (South African Government, 2013).

When using personal information to train machine-learning models, organisations must have a lawful justification for processing the data and ensure that it is adequate, relevant and not excessive for the intended purpose. They must also explain how the information will be used, protect it against unauthorised access and consider whether using it for a new purpose is compatible with the original purpose (Information Regulator South Africa, n.d.).

Personal information relates to an identifiable person. Examples include names, identity numbers, email addresses and financial histories. Anonymised information has been processed so that an individual can no longer reasonably be identified. Removing a person's name alone does not necessarily make information anonymous if other details can still identify them. Proper anonymisation can reduce privacy risks when using information for analytics and research.

Removing a person's name alone does not necessarily make information anonymous if other details can still identify them (South African Government, 2013).

Source:

1. South African Government (2013). Protection of Personal Information Act 4 of 2013. Available at: https://www.gov.za/documents/protection-personal-information-act

2. Information Regulator South Africa (n.d.). Eight conditions of lawful processing. Available at: https://inforegulator.org.za/wp-content/uploads/2020/07/20210727-InfoRegSA-ZoomWebinar-POPIA-presentation.pdf

AI tools declaration

I used ChatGPT and Claude AI as supporting tools while completing this assignment.

These tools were used to help explain machine-learning concepts, clarify Python code and error messages, and provide feedback on the accuracy and understanding of my responses.

I reviewed the information provided by these tools, made my own decisions about the content, and wrote the assignment in my own words. I also checked relevant technical information against official documentation, including the pandas and scikit-learn documentation.

The research, interpretation and final submitted work remain my responsibility.

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